Analysis and Observables

From a curve to three measurable constants, and the figure that carries the project

Phase 5 converts the scan into physics: equilibrium bond length, dissociation energy, and a vibrational frequency that infrared spectroscopy can check. Each extraction has a way of going quietly wrong, and this document covers all three.

It also confronts the result honestly. The VQE reproduces FCI to 0.0013 mHa and disagrees with experiment by 17%. Understanding why both statements are true, and are not in tension, is the point of the phase.

Estimated time: **2–3 hours**.

In this document

1. Where this phase sits

You have a curve. Phase 5 turns it into physics: three measurable constants, one publication-style figure, and the written analysis that is the actual deliverable of the project. It is also where the most instructive feature of the whole exercise finally becomes visible.

2. Theory

2.1 Fit the minimum — do not read it off

The lowest sampled point is not the minimum. It is the lowest point *you happened to evaluate*, and its position is quantised to your grid spacing. Reporting it means reporting your grid rather than the physics.

Fit a parabola through the few points nearest the bottom and solve for its vertex. For $E(R) \approx aR^2 + bR + c$ the minimum sits at:

$$R_e = -\frac{b}{2a}$$

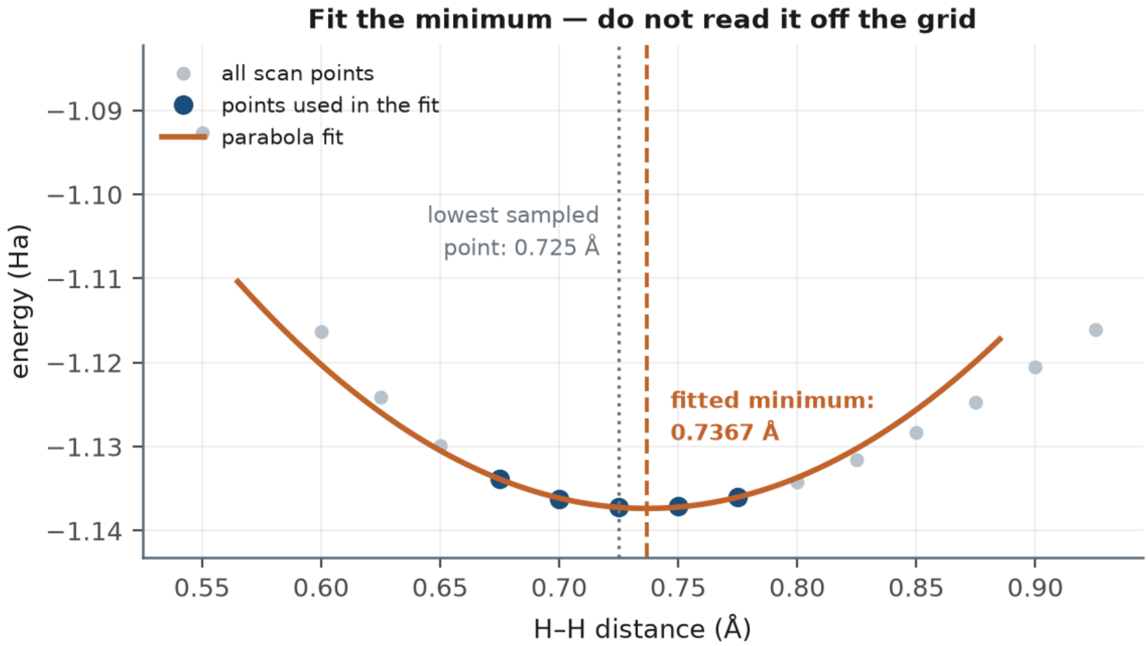

Figure 5.1 — The fitted minimum lands between grid points. The reference scan's lowest sample is at 0.7250 Å; the fit gives 0.7367 Å.

2.2 The dissociation limit is not your last grid point

A quiet error in the well depth

The obvious asymptote is the energy at the largest R you sampled. It is also wrong: at 2.5 Å the H_2 curve is **still bound by 2.9 mHa**. The molecule has not finished dissociating.

Use the analytic limit instead — two isolated hydrogen atoms in the same basis, computed directly. It costs one extra PySCF call and removes the error entirely.

$$E_\infty = 2\,E(\text{H atom, STO-3}G) = -0.933164\ \text{Ha}$$

The well depth is then the honest quantity:

$$D_e = E_\infty - E_{\text{min}}$$

2.3 Harmonic frequency: from curvature to spectroscopy

Near the bottom, any smooth well looks like a parabola, and a parabolic well is a harmonic oscillator. Its frequency is:

$$\omega = \sqrt{\frac{k}{\mu}}, \qquad k = \frac{d^2E}{dR^2}\bigg|_{R_e}, \qquad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

k	The **force constant** — the curvature at the minimum. A steep, narrow well means a stiff bond. From the parabola fit, k = 2a.
μ	The **reduced mass** of the two nuclei. For H_2, $\mu = m_H/2 \approx 918.7$ electron masses.
ω	The vibrational frequency, conventionally reported in wavenumbers (cm^{-1}).

This is the step that connects your curve to a real experiment. Infrared and Raman spectroscopy measure exactly this number, and for H_2 the experimental value is 4401.2 cm^{-1}. H_2 has the highest vibrational frequency of any molecule, because ω goes as $1/\sqrt{\mu}$ and hydrogen is the lightest atom there is.

Unit conversion is where this goes wrong

You fitted the parabola in Ångström, but $\omega = \sqrt{k/\mu}$ is an atomic-units formula. Convert the second derivative to Hartree per **Bohr** squared before dividing by the reduced mass:

```
k = (2 * coeffs[0]) / BOHR_PER_ANGSTROM**2
```

Skip this and your frequency is wrong by a factor of $1.8897^2 \approx 3.57$.

2.4 Anharmonicity: your fit window matters

A real potential well is not a parabola — it is steeper on the inside and flatter on the outside. That asymmetry is **anharmonicity**. Fit over too wide a window and the parabola absorbs some of it as if it were curvature, inflating your frequency.

The reference implementation prints a sensitivity table so you can see this directly rather than guess at it:

```
Fit-window sensitivity (anharmonicity check)
============================================================
   window   span/A     R_e/A  omega/cm-1
        2    0.100    0.7367      5184.1
        3    0.150    0.7390      5210.9
        4    0.200    0.7419      5246.3
        5    0.250    0.7456      5291.0
        6    0.325    0.7501      5593.2
        8    0.475    0.7606      6295.1
```

- The frequency drifts by **21%** across this range. Quoting a harmonic frequency without stating the fit window is meaningless.
- **Narrower is more correct.** A converged answer is one that stops changing as the window shrinks. Refitting on a dedicated fine grid (spacing 0.001 Å around the minimum) gives 4999 cm^{-1}, so even the narrowest window on the scan grid is still about 4% high.
- Note the trap in the other direction: the *wider* window gives R_e = 0.7419 Å, which is closer to the experimental 0.7414 Å than the correct 0.7349 Å. **A number that looks better can be more wrong** — here two errors happen to cancel.

Recommended practice

Use the scan grid for the curve and the figure. For the frequency, run a small dedicated calculation on a tight grid around the fitted minimum — three points at ±0.005 Å is enough, and it takes milliseconds classically.

2.5 The most instructive thing on your plot

Overlay the Hartree–Fock curve. As the atoms separate, restricted HF **turns upward and diverges** instead of flattening to the energy of two isolated hydrogen atoms. VQE and FCI flatten correctly.

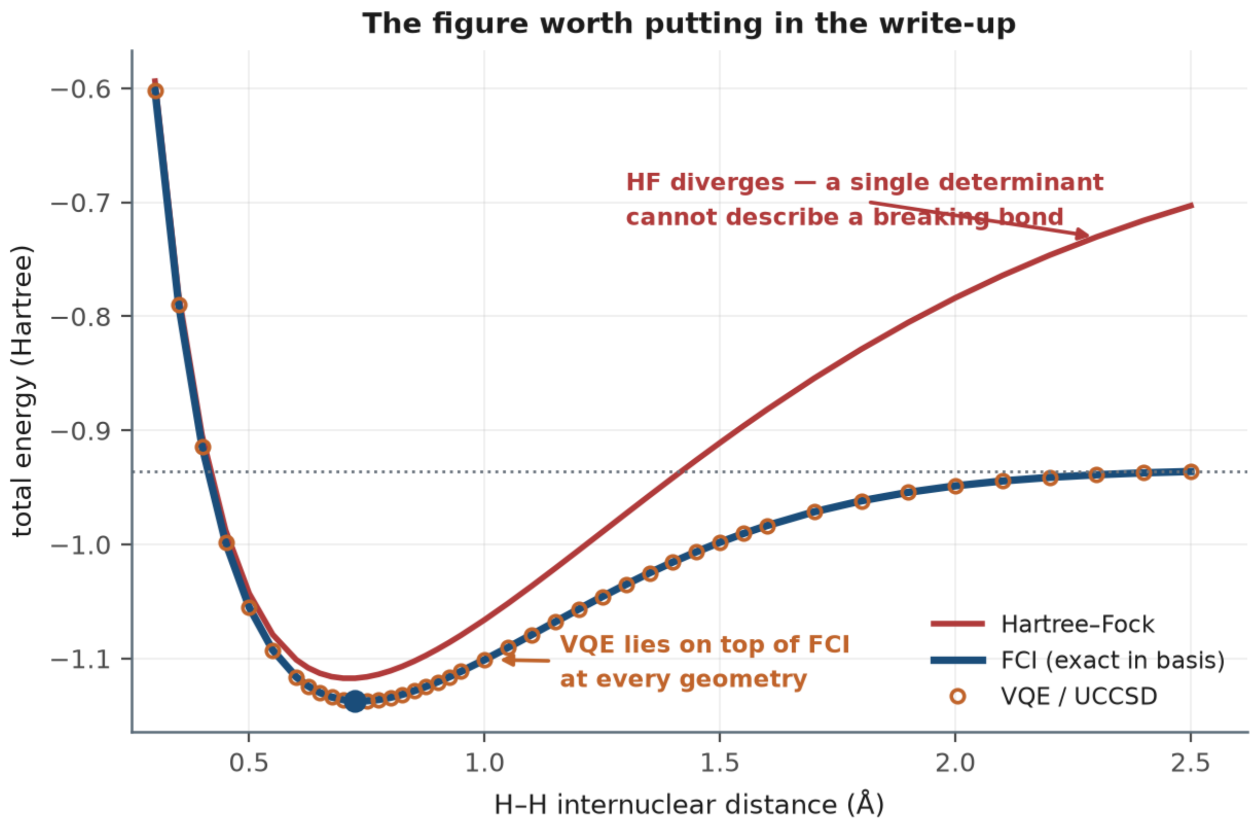

Figure 5.2 — The figure worth putting in your write-up. Real data from the reference implementation.

The reason is Phase 1's theory made visible. Restricted Hartree–Fock forces both electrons into the same spatial orbital, σ_g, which is delocalised over both atoms. At large separation that means each "atom" is described as a superposition of neutral H and ionic H^+/H^- — and the ionic component costs a great deal of energy that never goes away.

Correctly dissociated H_2 requires **two** determinants, $\sigma_g{}^2$ and $\sigma_u{}^2$, mixed with nearly equal weight. One determinant cannot do it, at any choice of orbitals. This is **static correlation**, and the visible gap between the red and blue curves at 2.5 Å is the clearest picture of it you will produce.

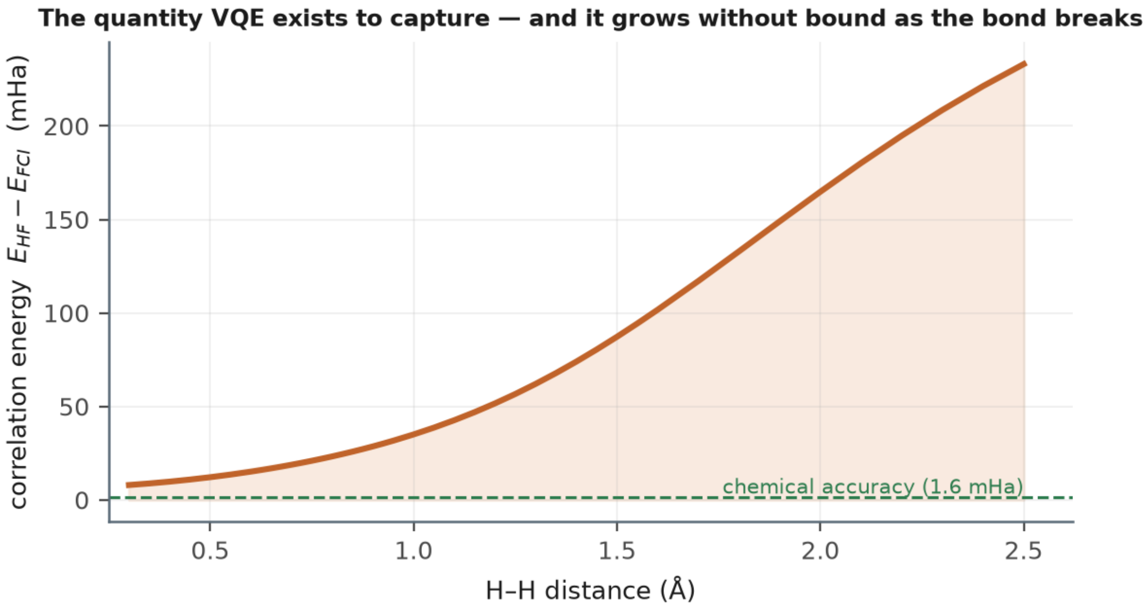

Figure 5.3 — The same fact plotted directly: correlation energy grows without bound as the bond breaks.

3. Build guide

Step 5.1 — Fitting and the dissociation limit

Listing 5.1 — fitting and the asymptote

```
"""analyse.py - extract physical observables from the scan and plot it."""

import matplotlib.pyplot as plt
import numpy as np
from pyscf import gto, scf

BOHR_PER_ANGSTROM = 1.8897261254578281
HARTREE_TO_EV = 27.211386245988
HARTREE_TO_CM = 219474.6313632
# atomic mass of 1-H in electron masses (1.00782503 u x 1822.888486 me/u)
MASS_H = 1837.4713
REDUCED_MASS_H2 = MASS_H / 2.0


def fit_minimum(r, e, window=2):
    """Locate the minimum by fitting a parabola, not by reading off the grid.

    `window` is the number of points either side of the lowest sampled
    point. SMALLER IS BETTER for the curvature: a real potential is
    anharmonic, so a wide window fits the anharmonicity as if it were
    curvature and inflates the frequency.
    """
    i = int(np.argmin(e))
    lo = max(0, i - window)
    hi = min(len(r), i + window + 1)

    coeffs = np.polyfit(r[lo:hi], e[lo:hi], 2)
    r_eq = -coeffs[1] / (2 * coeffs[0])
    e_min = np.polyval(coeffs, r_eq)
    return r_eq, e_min, coeffs


def dissociation_limit(basis="sto-3g"):
    """Energy of two infinitely separated H atoms, computed analytically.

    Do NOT use the last point of your scan as the asymptote. At 2.5 A the
    H2 curve is still bound by about 2.9 mHa, so the last grid point
    UNDERESTIMATES the well depth.
    """
    mol = gto.M(atom="H 0 0 0", basis=basis, spin=1, verbose=0)
    e_h = scf.UHF(mol).run(verbose=0).e_tot
    return 2.0 * e_h


def well_depth(e_min, e_asymptote):
    """Dissociation energy D_e in Hartree and eV."""
    d_e = e_asymptote - e_min
    return d_e, d_e * HARTREE_TO_EV
```

Step 5.2 — The frequency, with its unit conversion

Listing 5.2 — frequency and its convergence check

```
def harmonic_frequency(coeffs):
    """Vibrational frequency from the curvature at the minimum.
```

```
    The parabola was fitted in Angstrom, so the second derivative must be
    converted to Hartree per Bohr squared before combining with the reduced
    mass in atomic units.
    """
    d2E_dR2_angstrom = 2.0 * coeffs[0]                          # Ha / A^2
    k = d2E_dR2_angstrom / (BOHR_PER_ANGSTROM ** 2)             # Ha / Bohr^2

    omega_au = np.sqrt(k / REDUCED_MASS_H2)                     # atomic units
    return omega_au * HARTREE_TO_CM                             # cm^-1


def frequency_convergence(r, e, windows=(2, 3, 4, 5, 6, 8)):
    """Show how the extracted observables drift with the fit window.

    A converged answer is one that stops changing as the window shrinks.
    If your numbers are still moving at the narrowest window, your grid is
    too coarse near the minimum.
    """
    print(f"  {'window':>7} {'span/A':>8} {'R_e/A':>9} {'omega/cm-1':>11}")
    for w in windows:
        i = int(np.argmin(e))
        lo, hi = max(0, i - w), min(len(r), i + w + 1)
        if hi - lo < 3:
            continue
        _, _, co = fit_minimum(r, e, window=w)
        r_eq = -co[1] / (2 * co[0])
        print(f"  {w:>7} {r[hi-1]-r[lo]:8.3f} {r_eq:9.4f} "
              f"{harmonic_frequency(co):11.1f}")
```

Step 5.3 — Run it

Actual output from the reference implementation

```
$ python analyse.py
Extracted from the VQE curve
============================================================
  equilibrium bond length       0.7367 A     (exp 0.7414, -0.63 %)
      lowest grid point         0.7250 A
  minimum energy             -1.137349 Ha
  dissociation limit 2E(H)   -0.933164 Ha
      last grid point        -0.936055 Ha  (still bound by -2.89 mHa)
  well depth D_e                5.5562 eV    (exp 4.7466, +17.1 %)
  harmonic frequency            5184.1 cm-1  (exp 4401.2, +17.8 %)

  max |VQE - FCI|             0.001257 mHa
  correlation energy at R_e    20.0062 mHa
```

Step 5.4 — Interpret the disagreement with experiment

Quantity	This work	Experiment	Error	Verdict
Bond length R_e	0.7367 Å	0.7414 Å	−0.63%	Good — minimal basis gets geometry roughly right
Well depth D_e	5.556 eV	4.747 eV	+17.1%	Poor — STO-3G substantially overbinds
Frequency ω	5184 cm^{-1}	4401 cm^{-1}	+17.8%	Poor — well too deep and too narrow, so the bond is too stiff

This disagreement is not a bug — and saying so is the point

A minimal basis is a deliberately impoverished description of the electronic structure. It gives poor absolute energies and poor energy *differences*, and the two errors above are the expected consequence.

Your VQE reproduced FCI to 0.0013 mHa. That is the claim the project supports, and it is completely independent of how well FCI/STO-3G matches reality. The quantum algorithm did its job perfectly; the basis set is what limits the physics.

Stating both facts clearly — "exact agreement with FCI, 17% disagreement with experiment, and here is why those are consistent" — demonstrates that you understand what you built. Quietly reporting only the first, or apologising for the second, does not.

If you want the numbers to match experiment

Change the basis, not the algorithm. Re-run the whole pipeline with `basis="6-31g"` or `"cc-pvdz"` — every function you wrote takes basis as an argument, so this is a one-line change. The qubit count rises (cc-pVDZ gives 20 spin orbitals, so 20 qubits) and the state-vector simulation gets slow, which is itself an instructive encounter with why quantum chemistry is hard.

4. The figure

One plot, three curves, minimum marked, asymptote annotated. Do not clutter it. The things that must be legible are: that VQE and FCI are indistinguishable, that Hartree–Fock diverges, and where the minimum is.

Listing 5.3 — the figure

```
def plot(path="data/scan.npz",
         out="results/figures/dissociation_curve.png", window=2):
    d = np.load(path)
    r, e_hf, e_fci, e_vqe = d["grid"], d["e_hf"], d["e_fci"], d["e_vqe"]
    r_eq, e_min, _ = fit_minimum(r, e_vqe, window=window)
    e_inf = dissociation_limit()

    fig, ax = plt.subplots(figsize=(7, 4.5))
    ax.plot(r, e_hf, "-", color="#b03a3a", lw=2, label="Hartree-Fock")
    ax.plot(r, e_fci, "-", color="#1a4d7a", lw=2.5, label="FCI (exact)")
    ax.plot(r, e_vqe, "o", color="#c0632a", ms=4.5, mfc="none", mew=1.3,
            label="VQE (UCCSD)")
    ax.axhline(e_inf, color="grey", ls=":", lw=1.2)
    ax.text(2.45, e_inf + 0.008, "2 x E(H)", fontsize=8, color="grey",
            ha="right")
    ax.plot([r_eq], [e_min], "*", color="#1a4d7a", ms=15, zorder=5)
    ax.annotate("", xy=(2.1, e_min), xytext=(2.1, e_inf),
                arrowprops=dict(arrowstyle="<|-|>", color="#2e7d4f", lw=1.4))
    ax.text(2.15, (e_min + e_inf) / 2, "$D_e$", color="#2e7d4f", fontsize=10,
            va="center")

    ax.set_xlabel("H-H internuclear distance (Angstrom)")
    ax.set_ylabel("total energy (Hartree)")
    ax.set_title("H$_2$ dissociation curve, STO-3G")
    ax.legend(frameon=False, loc="lower right")
    ax.grid(alpha=0.25)
    fig.tight_layout()
    os.makedirs(os.path.dirname(out), exist_ok=True)
    fig.savefig(out, dpi=200)
```

A second figure is worth making for the write-up: the error $|E_{VQE} - E_{FCI}|$ on a logarithmic axis, with the chemical-accuracy threshold drawn in. It is the plot that demonstrates verification rather than merely asserting it.

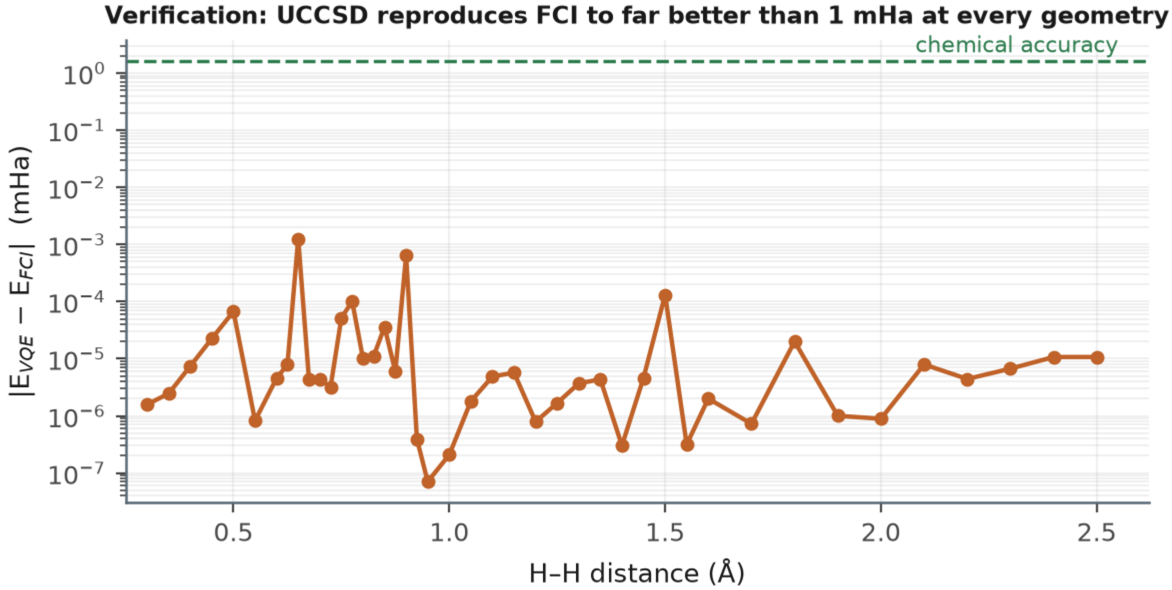

Figure 5.4 — Verification made visible. UCCSD sits six orders of magnitude below chemical accuracy at every geometry.

5. Writing it up

The write-up is the deliverable. Four short sections are enough:

1. **What was computed.** H_2 dissociation curve, STO-3G, 43 geometries, VQE with UCCSD against Hartree–Fock and FCI references.
2. **Verification.** Maximum deviation from FCI across the whole curve: 0.0013 mHa. Include the log-scale error plot. State that UCCSD is exact for this system, so this is the expected result and its absence would indicate a bug.
3. **Physics extracted.** The three constants, each with its experimental comparison and an explanation of the disagreement. Include the fit-window sensitivity — it shows you checked.
4. **Where Hartree–Fock fails and why.** The divergence at large R, the single-determinant argument, and static correlation. This is the section that shows understanding rather than execution.

6. What was done in this phase

- **Fitted the minimum rather than reading it off**, removing the grid-spacing quantisation from the reported bond length.
- **Used the analytic dissociation limit** instead of the last grid point, having established that the curve is still bound by 2.9 mHa at 2.5 Å.
- **Extracted a harmonic frequency** from the curvature and the reduced mass, including the Ångström-to-Bohr conversion that otherwise costs a factor of 3.57.
- **Quantified anharmonicity** by measuring how the extracted observables drift with fit window — a 21% spread in ω — and identified that a number closer to experiment can be the more wrong number.

- **Produced the three-curve figure** and the log-scale error plot: one showing the physics, one showing the verification.
- **Explained the Hartree–Fock divergence** as static correlation — the single-determinant approximation failing qualitatively rather than quantitatively — which is the central physical insight of the project.
- **Separated two independent claims**: exact agreement with FCI (the algorithm worked) and 17% disagreement with experiment (the basis set is minimal). Both are true and neither undermines the other.

7. Checklist before moving on

1. Fitted R_e lies between grid points, not on one.
2. Well depth uses 2×E(H), not the last scan point.
3. The frequency calculation converts Å to Bohr.
4. You have run the fit-window sensitivity check and can state which window you used and why.
5. The three-curve figure shows HF diverging while VQE and FCI flatten.
6. You can explain in two sentences why restricted Hartree–Fock cannot dissociate H_2 correctly.

Next

Phase 6 — The extension. Phases 0–5 are a complete, verified project. Phase 6 is what makes it *yours*: a genuine experiment with a result that is not in the tutorials, built from machinery you already have.